

The Purple-Pipe Loop

Ladera Ranch's runoff-recycling system, mapped, audited, and now ROUTE-RESOLVED · LEHRP · v2, 11 Sept 2026

Independent research project; not medical advice; establishes no causation. No contaminant is asserted present in any water or soil discussed; the relevant streams have not been publicly tested for the constituents examined — itself a central finding. Statement classes marked throughout.

1 · The loop is real — verified (A2)

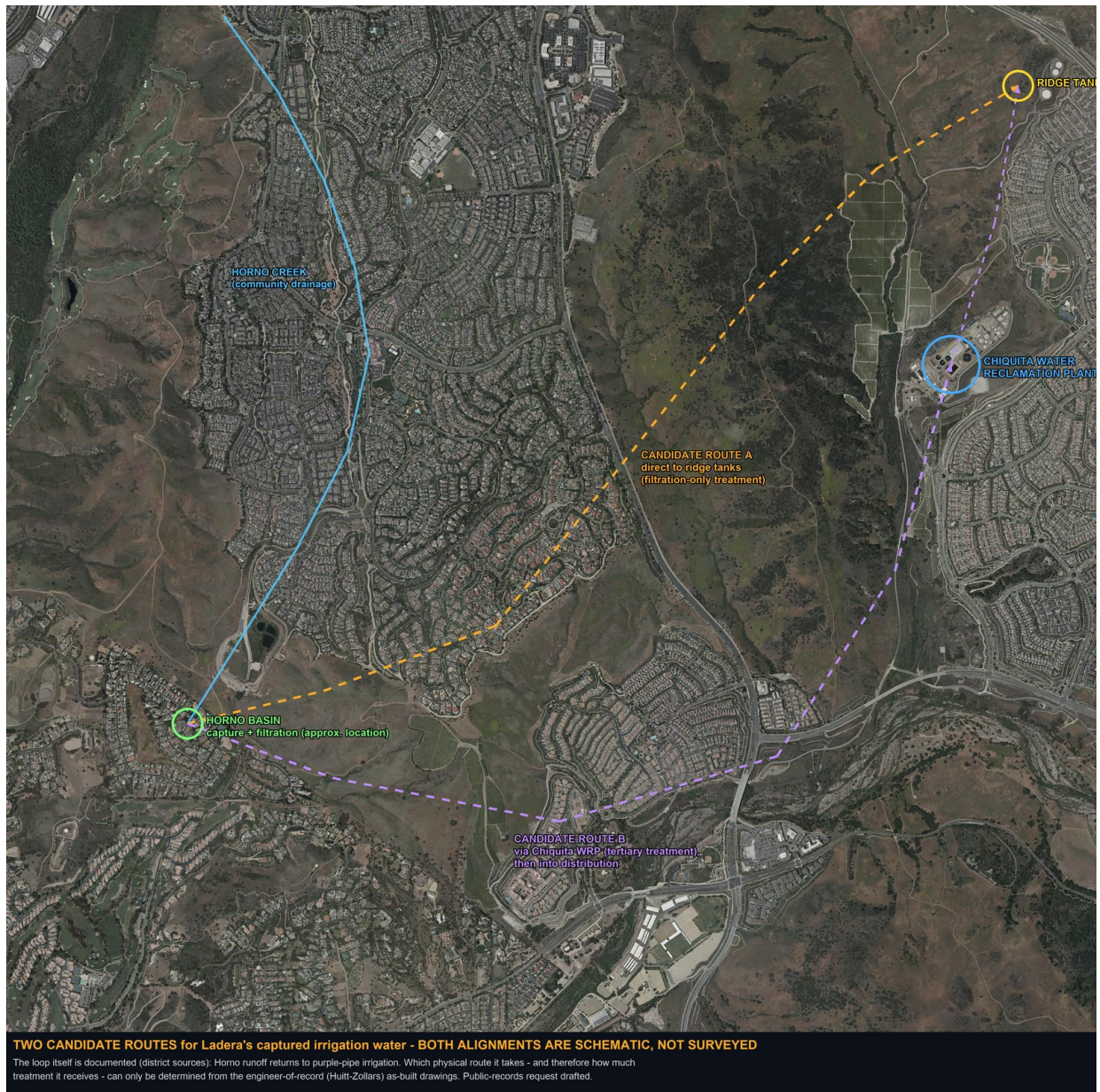
District sources: it "collects urban return (runoff) water from Oso Creek, Dove Creek, and **Horno Creek to blend with the recycled water... for irrigation**"; the "**Horno Basin...** designed to capture, treat and deliver millions of gallons of runoff water **for local irrigation use.**" Delivery is via the purple-pipe (recycled, non-potable) system irrigating Ladera's slopes, parks, medians and common-area turf.

2 · The route — RESOLVED from the regulatory record (A2, held with SHA-256)

San Diego Regional Water Board, **Addendum No. 4 to Order 97-52** (adopted 8 Oct 2008), permitting the **Horno Urban Water Recovery Facility** (Tract 15793, southern Ladera Ranch): runoff is collected in the **Horno Water Quality Basin — owned and operated by the County of Orange** — then flows "**through seven acres of constructed wetlands, owned by Ladera Ranch Homeowners Association, prior to blending with recycled water,**" entering the district's recycled distribution "**for subsequent landscape irrigation use**" — about **180 acre-feet (~59 million gallons) per dry season.**

The captured runoff does not pass through the Chiquita treatment plant. Its entire treatment train is the county basin plus the HOA wetlands — effective for sediment, trash and nutrients; not engineered for dissolved metals, pesticides, or pharmaceuticals.

What the permit measured: characterization sampling (Nov 2004–Apr 2005, point of diversion) reports salts and nutrients only — TDS 686 mg/L, sulfate 349, sodium 57, chloride 48.5, boron 0.22, nitrate 0.6, fluoride 0.275, phosphorus 0.4. **No arsenic, pesticide, or pharmaceutical results appear anywhere in the addendum.** Ongoing monitoring references the 1997 base order's Table A-1 list (base order obtainable — arsenic inclusion checkable). The facility was permitted via CEQA **Notice of Exemption** (29 Sept 2006): no environmental document analyzed the diverted stream's chemistry.

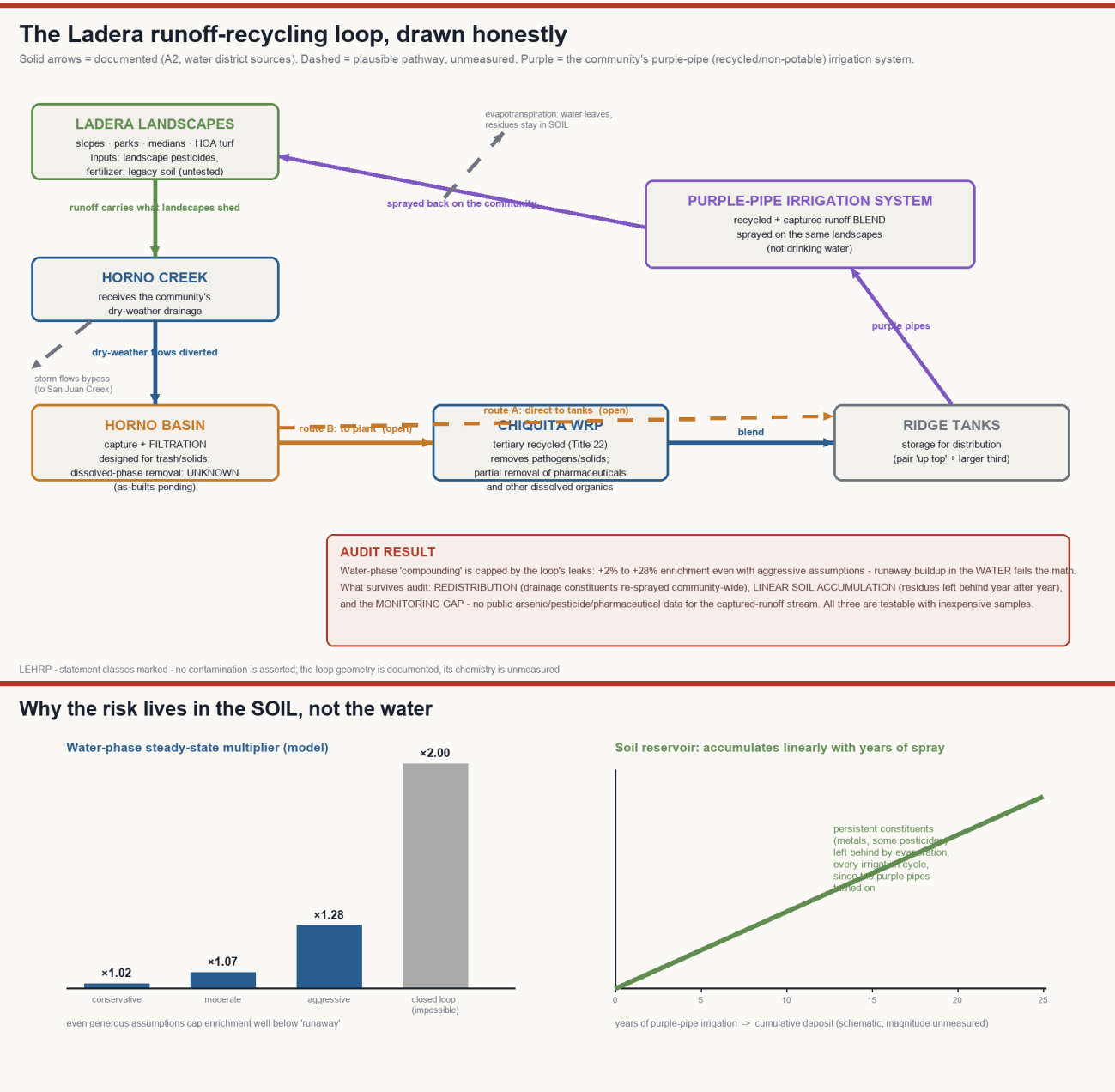


Resolved: direct-to-distribution (amber, in substance) with basin+wetlands as sole treatment; the plant route (purple) is ruled out for this stream. Alignments schematic; destination and treatment now documented.

3 · The audit — what fails, what survives

FAILS: water-phase "compounding." The loop is partial (storm bypass, evapotranspiration, blending dilution): modeled enrichment caps at $\times 1.02$ – $\times 1.28$ even aggressively. Runaway buildup in the water is not physically supported.

SURVIVES: (1) REDISTRIBUTION — whatever the drainage carries is re-sprayed community-wide through the purple pipes; (2) SOIL ACCUMULATION — evaporation leaves persistent constituents (metals incl. arsenic, persistent pesticides) in landscape soils, accumulating linearly across two-plus decades; (3) the MONITORING GAP — now sharpened by the permit itself: the captured-runoff stream was characterized for salts and nutrients only, and its treatment train includes no engineered removal of dissolved metals or trace organics.



4 · Verdict

Not supported as stated: "one big ongoing toxic loop that compounds" — fails mass balance. **Supported, restated defensibly:** Ladera Ranch operates a documented capture-and-respray loop: county basin → HOA wetlands → blended into the purple pipes, bypassing the treatment plant, ~180 AF per dry season. The loop redistributes whatever the drainage carries, deposits persistent constituents into landscape soils year over year, and its permit record shows characterization for salts and nutrients only — no arsenic, pesticide, or pharmaceutical data — on land where a federal arsenical dipping program operated a century ago and where soil has never been tested for it. The reported pattern warrants investigation; current data are insufficient to determine whether any constituent is present at levels of concern.

5 · What settles it — four inexpensive samples

1) Irrigation water at a purple-pipe sprinkler head (arsenic + speciation, pesticide panel, CEC screen). 2) Horno Basin sediment core. 3) Long-irrigated turf/slope soil vs never-irrigated control. 4) Wetland-outlet/blend-point water. Records worth having anyway: base Order 97-52 Table A-1 (does compliance monitoring include arsenic?); as-builts for exact alignment (three-party records: County OCPW - basin; LARMAC HOA - wetlands; SMWD - diversion works).

Sources: RWQCB Addendum No. 4 to Order 97-52 (held, SHA-256 97479faa...d5771) · district recycled-water page + May 2012 newsletter (archived) · county aerial imagery 2025 · regional literature. LEHRP · corpus timestamped under version control.